Robot-Guided Home Rehabilitation

Mid-term: setup, architecture and proof of concept

Team 105 · Leonid Pomortsev · Matin Mahmoudzadeh

Problem

After a stroke or surgery, patients practise walking **at home, alone.**

Concept

A robot that supervises walking exercise at home.

At home today	With our robot
Loses the rhythm	Plays a beat to walk, counts the steps
Gets tired, stops early	Notices it, gives a short correction
The physio never sees it	Sends the physio a summary

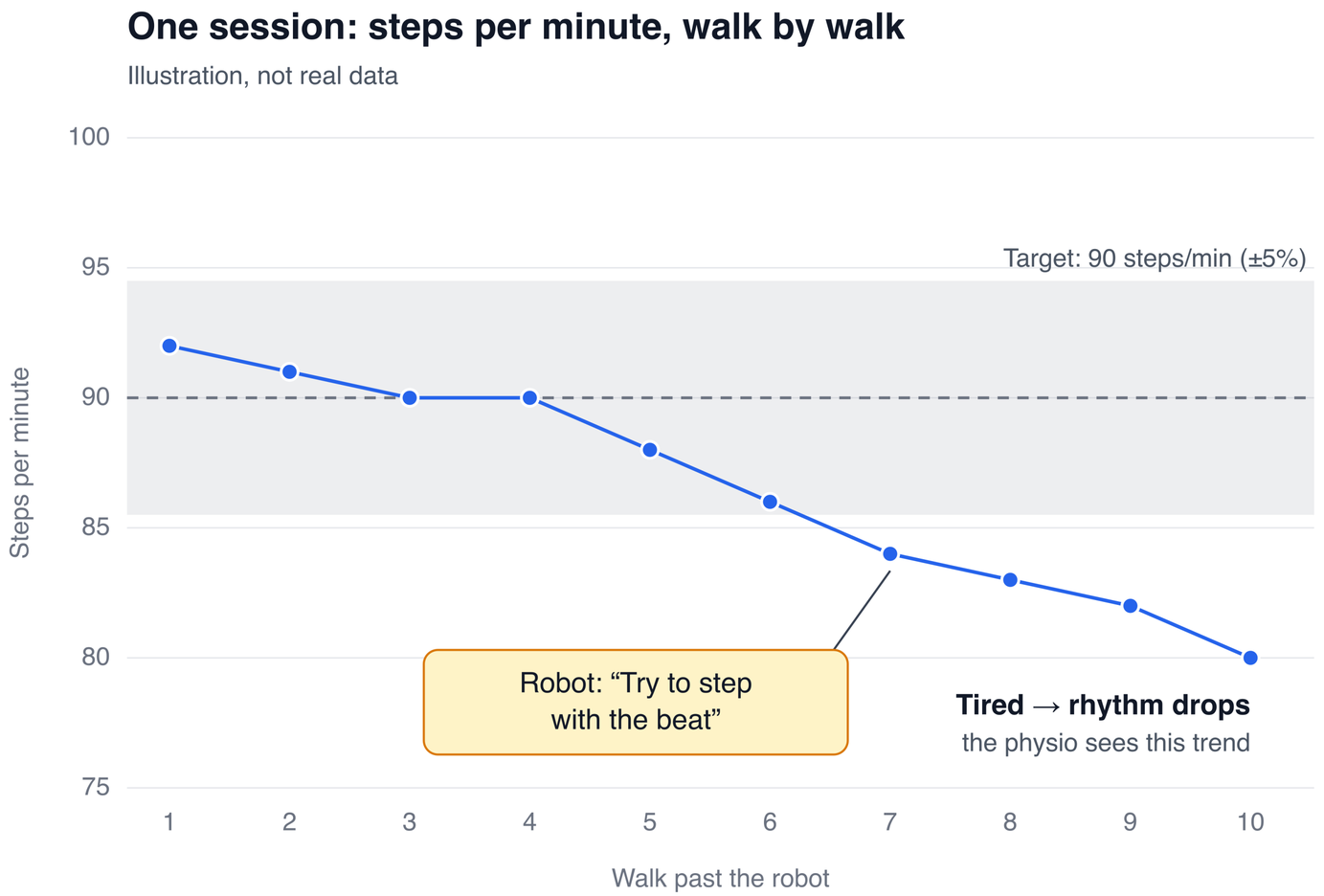
The robot supports the physio between appointments; it doesn’t replace them.

Why Cadence?

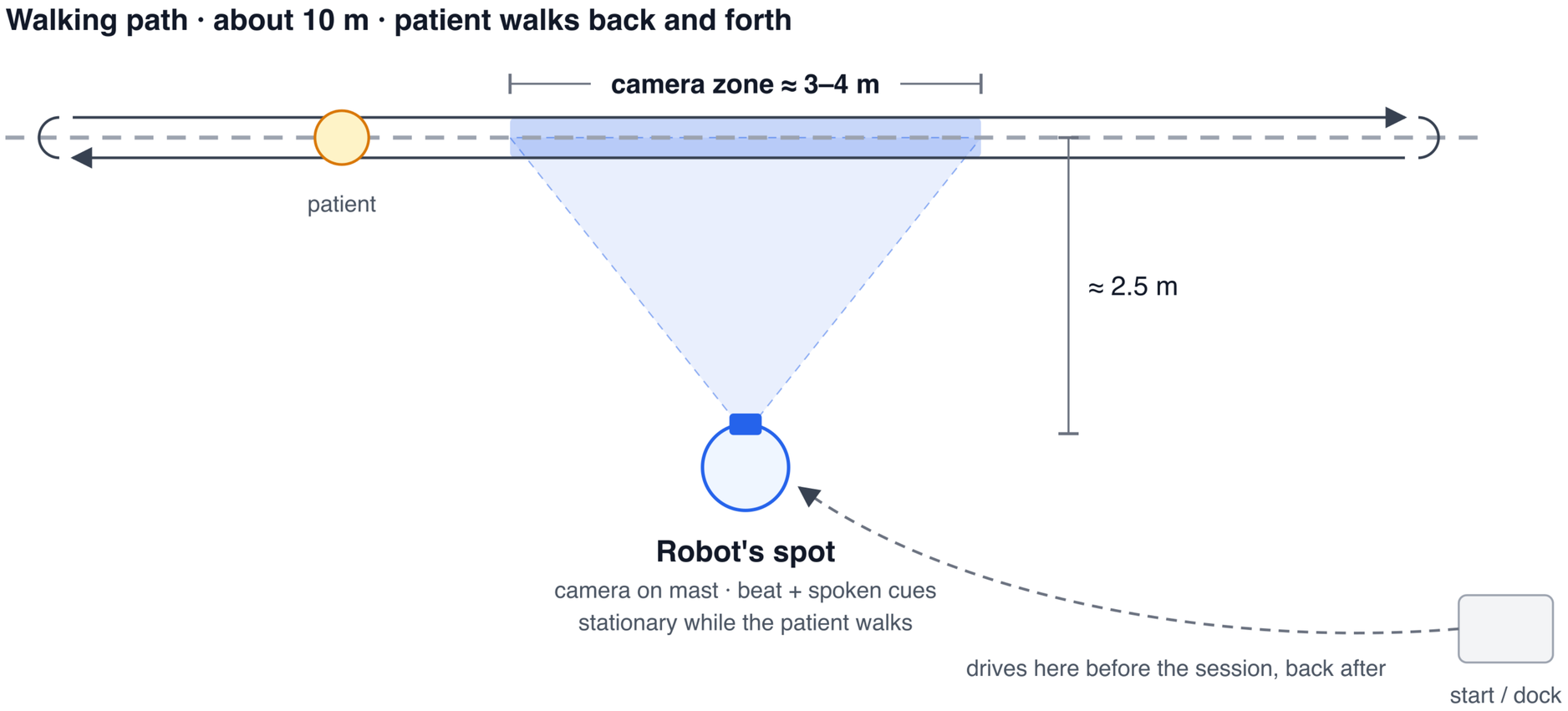
Cadence = steps per minute

- The physio sets a target cadence; the robot checks whether the patient keeps it.
- Walking to a beat helps gait after a stroke and in Parkinson's.*
- Used twice: **during** the walk to correct, **after** as a trend.

Takeaway: everything depends on **counting steps correctly.**



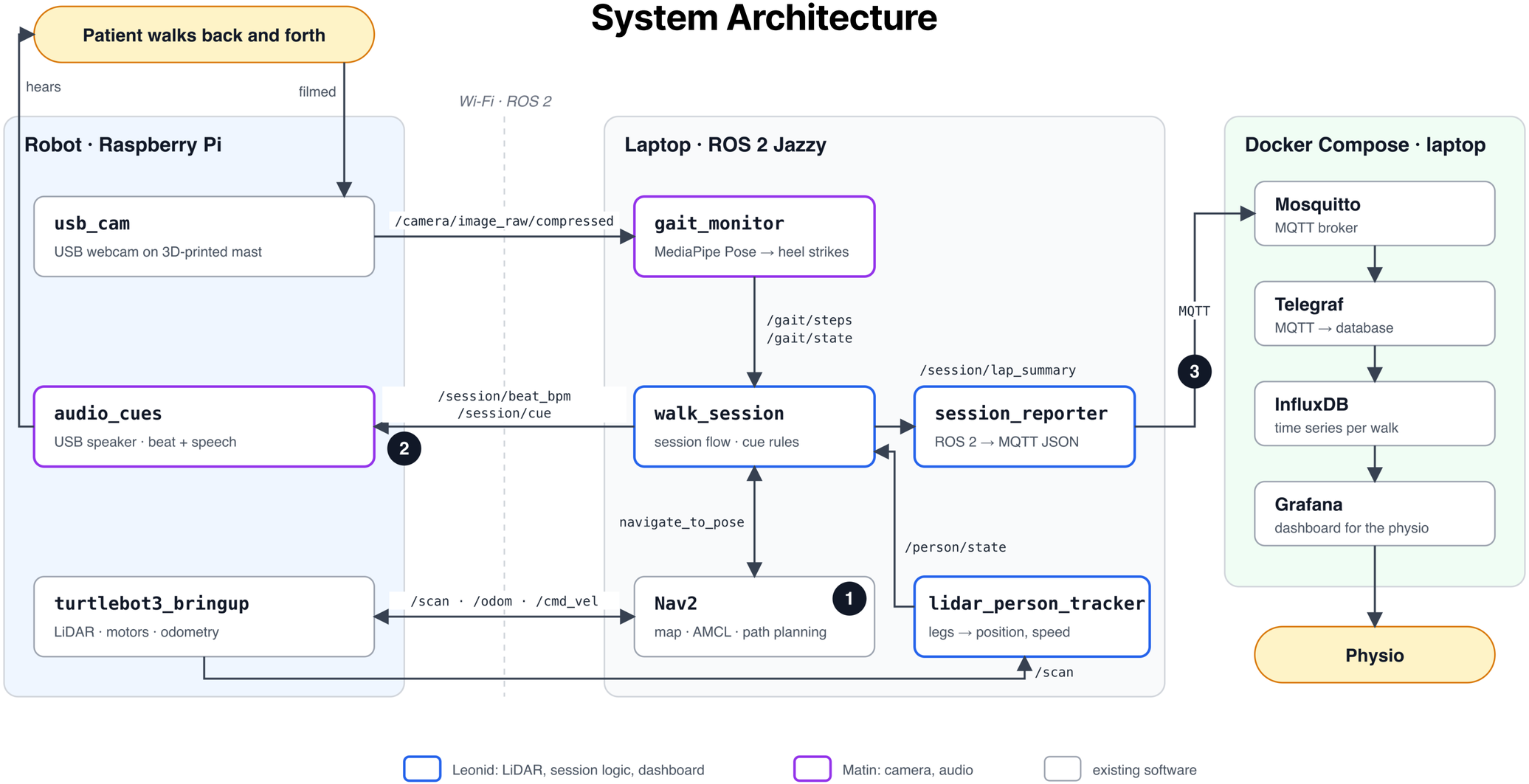
*Systematic reviews: Ghai et al., Sci. Rep. 2018 (Parkinson's); Ghai & Ghai, Sci. Rep. 2019 (stroke)



Each walk through the camera zone is one measurement. Distances approximate; robot not to scale.

Physical Setup

System Architecture



1. **Getting to its spot:** a map of the room and standard ROS 2 navigation
2. **Talking to the patient:** the beat is generated on the robot itself, so Wi-Fi delays can't disturb it
3. **Reporting to the physio:** after each walk, a short message goes to the dashboard

Step Detection & Sensor Fusion Pipeline

